

# SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence      COURSE CODE: CSA17

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## COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

**Assessment Tool Weightage: 50%**

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|----------------------|--------------------------|
| <b>Minutes</b>       | <b>Time Duration: 30</b> |
| <b>QUESTIONS: 30</b> | <b>Total Marks:30</b>    |
| <b>Annexure A</b>    |                          |
| <b>MCQ</b>           |                          |

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## Code of Conduct:

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

# Annexure A

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
- b) An agent that uses a knowledge base and logical inference to decide actions
- c) An agent that only follows pre-defined scripts
- d) An agent that learns from rewards

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a)  $P \wedge \neg Q$
- b)  $\neg P \vee Q$
- c)  $\neg Q \rightarrow \neg P$  only
- d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
- b) A conjunction of disjunctions (clauses)
- c) A conjunction of literals only
- d) A sequence of implications

Q4. In FOL, the sentence  $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$  means:

- a) Some students pass
- b) Every entity that is a student passes
- c) If something passes, it is a student
- d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

6. What is the core function of a logical agent in AI?

- a) Reacting only to current percepts
- b) Using logic and knowledge representation to reason and make decisions
- c) Performing physical actions without internal representation
- d) Relying solely on sensors for navigation

7. Which component of a logical agent is responsible for storing facts and rules about the world?

- a) Inference Engine
- b) Perception Unit
- c) Knowledge Base (KB)
- d) Action Selector

8. What does a knowledge-based agent do immediately after obtaining a percept from the environment?

- a) Selects an action
- b) Executes the action
- c) Updates the Knowledge Base with the new percept

d) Reasons to derive new information

9. Which operation is used to add new information to a Knowledge Base?

a) ASK b) TELL c) PERCEIVE d) UPDATE

10. What is a proposition in formal logic?

- a) A command or request
- b) A declarative statement that is either true or false
- c) A question regarding the environment
- d) A letter representing a variable

11. Which of the following is NOT a proposition?

- a) "The sun rises in the east."
- b) " $5 + 3 = 10$ ."
- c) "Close the door."
- d) "Chennai is in India."

12. Which logical operator is represented by the symbol  $\neg$ ?

- a) AND
- b) OR
- c) NOT
- d) Implication

13. The AND ( $\wedge$ ) operator is true only when:

- a) At least one proposition is true
- b) Both propositions are true
- c) Both propositions have different truth values
- d) Neither proposition is true

14. A proposition that is always true, regardless of the truth values of its variables, is called a:

- a) Contradiction b) Tautology c) Contingency d) Equivalence

15. In an implication ( $P \rightarrow Q$ ), the statement is false only when:

- a) P is false and Q is true
- b) P is true and Q is false
- c) Both P and Q are false
- d) Both P and Q are true

16. What is the number of rows in a truth table for a logical expression with 3 propositions?

- a) 4 b) 6 c) 8 d) 9

17. Which of these represents one of De Morgan's Laws?

- a)  $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- b)  $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$

- c)  $\neg(\neg P) \equiv P$
- d)  $P \vee \neg P \equiv \text{True}$

18. A proposition that is sometimes true and sometimes false is known as a:

- a) Tautology b) Contradiction c) Contingency d) Implication

19. Which logical connective is true if and only if both propositions have the same truth value?

- a) Implication ( $\rightarrow$ ) b) Biconditional ( $\leftrightarrow$ ) c) OR ( $\vee$ ) d) NOT ( $\neg$ )

20. According to operator precedence, which operator should be evaluated first?

- a) AND ( $\wedge$ ) b) OR ( $\vee$ ) c) NOT ( $\neg$ ) d) Implication ( $\rightarrow$ )

21. What is the process of deriving a conclusion from given premises called?

- a) Perception b) Representation c) Inference d) Predication

22. Which inference rule follows the form: If P, then Q; P; Therefore, Q?

- a) Modus Tollens b) Modus Ponens c) Hypothetical Syllogism d) Resolution

23. Which valid reasoning pattern is represented by:  $P \rightarrow Q$ ;  $\neg Q$ ; Therefore,  $\neg P$ ?

- a) Modus Ponens b) Disjunctive Syllogism c) Modus Tollens d) Addition

24. What can be concluded from the premises  $(P \rightarrow Q)$  and  $(Q \rightarrow R)$  using Hypothetical Syllogism?

- a)  $P \wedge R$  b)  $P \vee R$  c)  $P \rightarrow R$  d)  $\neg P \rightarrow \neg R$

25. The rule that states if "P or Q" is true and "Not P" is true, then "Q" must be true is called:

- a) Simplification b) Conjunction c) Disjunctive Syllogism d) Resolution

26. Which rule allows you to conclude "P" from the premise "P and Q"?

- a) Addition b) Simplification c) Conjunction d) Modus Ponens

27. The inference rule Resolution follows which symbolic form?

- a)  $P \therefore P \vee Q$  b)  $P, Q \therefore P \wedge Q$  c)  $P \vee Q, \neg P \vee R \therefore Q \vee R$  d)  $P \rightarrow Q, Q \rightarrow R \therefore P \rightarrow R$

28. Which of the following is an example of the "Addition" rule?

- a) P and Q, therefore P
- b) P, therefore P or Q
- c) P, Q, therefore P and Q
- d) P or Q, not P, therefore Q

29. "If it rains, the ground is wet. The ground is wet. Therefore, it rained." This is an example of which invalid reasoning pattern?

- a) Denying the Antecedent b) Affirming the Consequent c) Modus Ponens d) Double Negation

30. What is the name of the invalid pattern:  $P \rightarrow Q$ ;  $\neg P$ ; Therefore,  $\neg Q$ ?

a) Affirming the Consequent b) Denying the Antecedent c) Modus Tollens d) Hypothetical Syllogism

## RUBRICS FOR CALCULATION

| Criteria                 | Weightage (%) | Level 4 (Exemplary)                                                   | Level 3 (Proficient)                         | Level 2 (Developing)                               | Level 1 (Beginning)                                |
|--------------------------|---------------|-----------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| Conceptual Understanding | 30%           | Demonstrates thorough understanding; answers all questions accurately | Shows good understanding with few errors     | Partial understanding; several incorrect responses | Lacks understanding; majority of answers incorrect |
| Accuracy of Answers      | 25%           | All answers are correct                                               | Most answers are correct with minor mistakes | Some correct answers; frequent errors              | Mostly incorrect answers                           |
| Application of Knowledge | 20%           | Correctly applies concepts to problem-based questions                 | Applies concepts with minor errors           | Limited ability to apply concepts                  | Unable to apply concepts                           |
| Time Management          | 15%           | Completes quiz well within time efficiently                           | Completes on time with minor delays          | Requires extra time or rushes through questions    | Unable to complete within given time               |
| Question Interpretation  | 10%           | Interprets all questions correctly and responds appropriately         | Misinterprets a few questions                | Often misinterprets questions                      | Unable to understand questions                     |

12-8-26

CO3-AT2 MCQ Test

M. Sravya  
192472192

- 1) B ✓
- 2) B ✓
- 3) B ✓
- 4) B ✓
- 5) B ✓
- 6) B ✓
- 7) C ✓
- 8) C ✓
- 9) B ✓
- 10) B ✓
- 11) B ✓
- 12) C ✓
- 13) A ✓
- 14) B ✓
- 15) B ✓
- 16) D ✓
- 17) A ✓
- 18) C ✓
- 19) B ✓
- 20) A ✓
- 21) B ✓
- 22) B ✓
- 23) A ✓
- 24) C ✓
- 25) B ✓
- 26) A ✓
- 27) C ✓
- 28) C ✓
- 29) B ✓
- 30) C ✓

18x1  
30